

the value of
information

CSCI
270

you are a vampire
with a taste for
AB blood

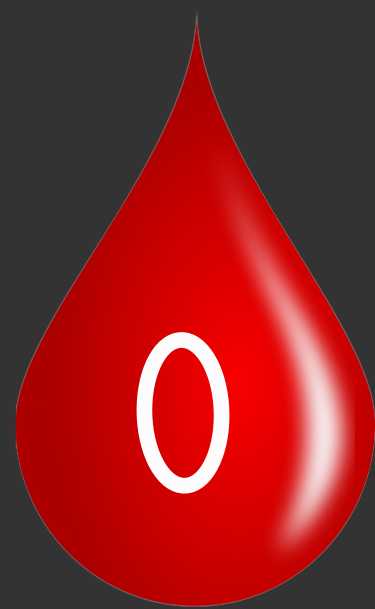
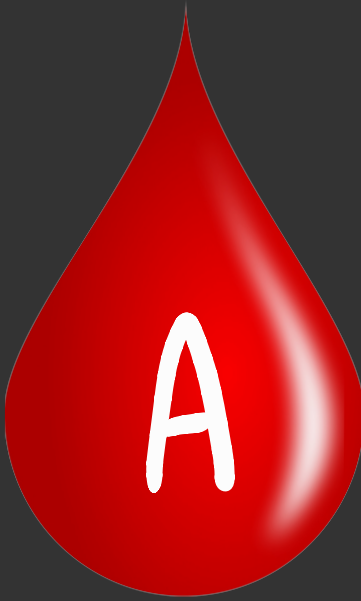
you have an unscrupulous
contact at the local
hospital who provides
you patient information
... for a price



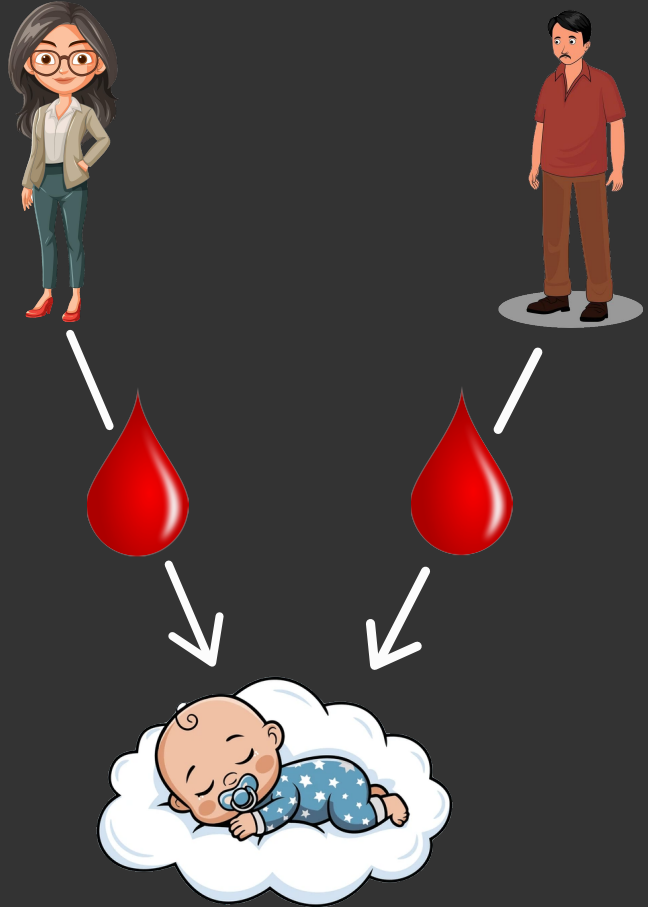
a short
primer on
blood



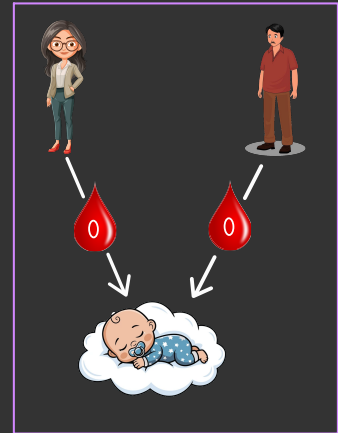
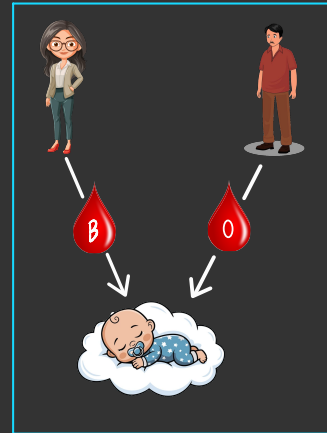
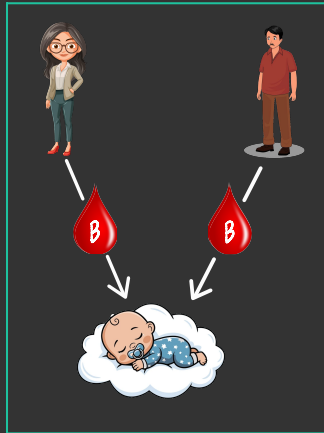
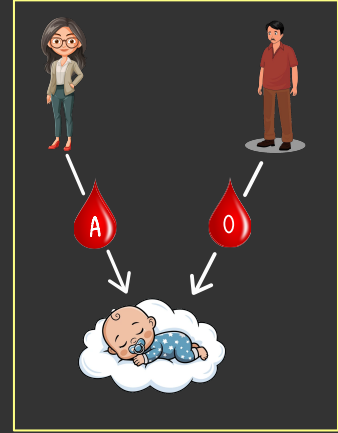
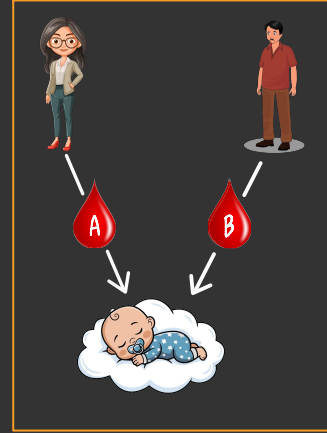
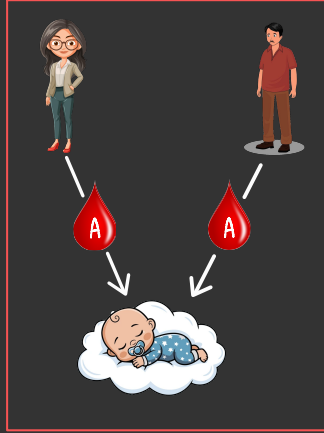
there are 3 blood type genes



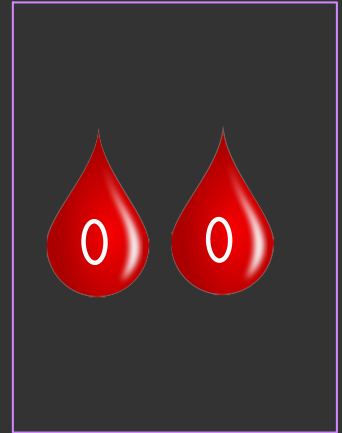
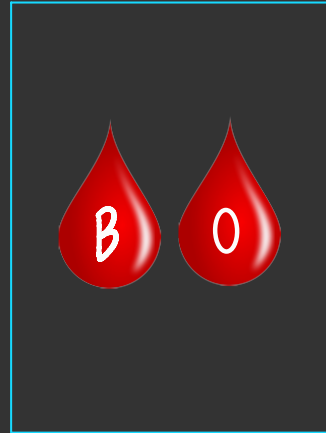
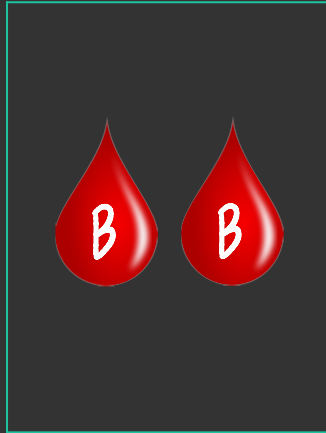
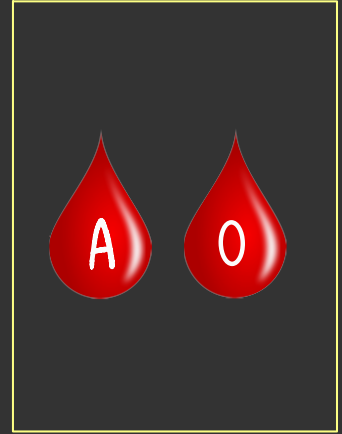
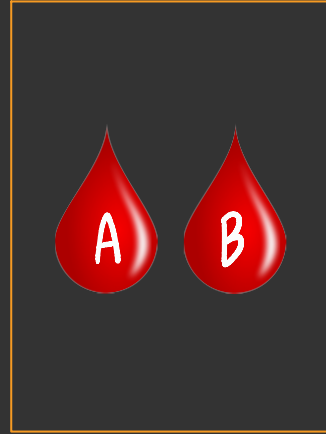
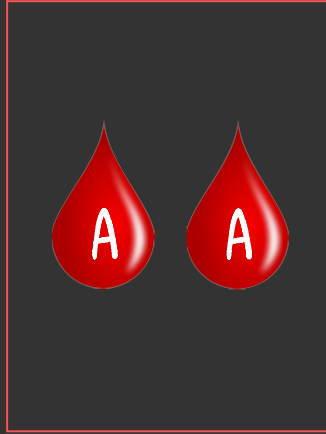
a person inherits
one gene
from each of
their parents



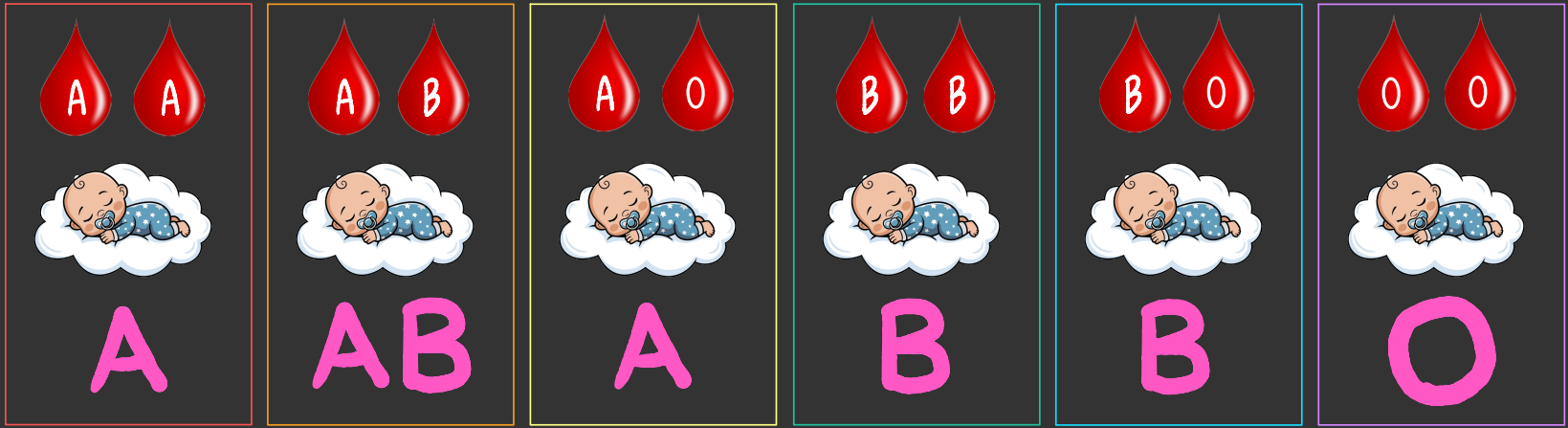
this can result
in six different
genotypes



this can result
in six different
genotypes



these hidden genetics manifest as
one of four observable **phenotypes**



anyways

as previously mentioned
you are a vampire
with a taste for

AB blood



you are starting
to get the sense that
your neighbor Yves
has "AB energy"



Yves

unfortunately Yves has
never had his blood drawn
at the local hospital, so
your hospital contact does
not know his blood type



Yves

but your contact can
offer you the blood
type of Yves' wife Xena



Xena



Yves

should you buy it?



no.

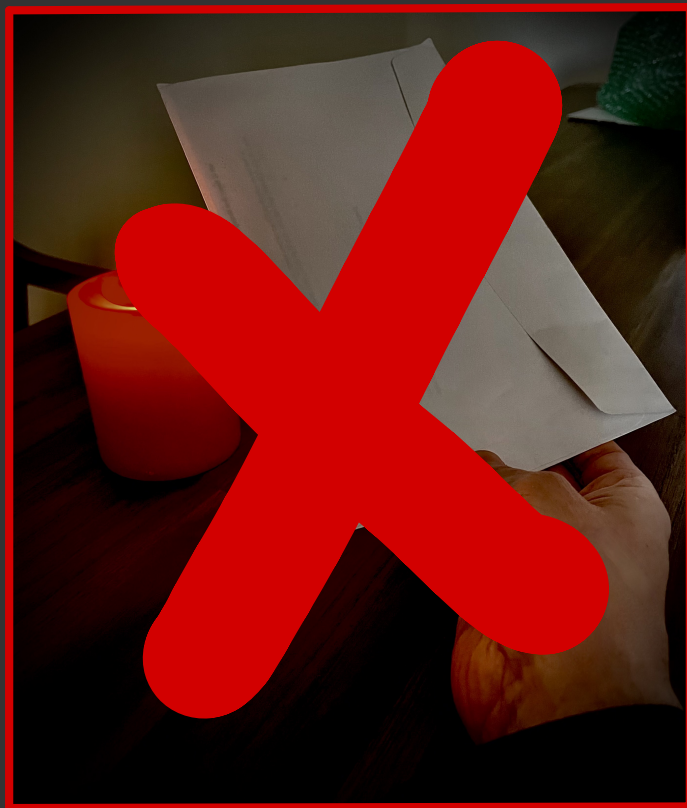


Xena



Yves

they're not
blood relatives



undaunted, your contact
returns with the blood
type of Zelda, the
daughter of Xena and Yves



Xena



Zelda



Yves

should you buy that?



yes.



Xena



Zelda



Yves

it could give us
information about
Yves' blood type



yes.



e.g.

if Zelda is type O,
Yves cannot be AB

it could give us
information about
Yves' blood type

turns out Zelda
is type A



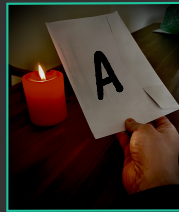
Xena



Zelda



Yves



at this point, your
contact reminds you
that Xena's blood
type is still available
for purchase

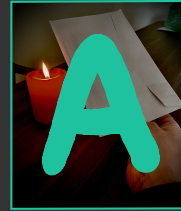
should you
buy it now?



Xena



Zelda



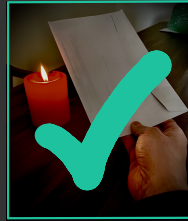
Yves

yes.

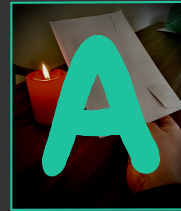
now it **could** give us
information about
Yves' blood type



Xena



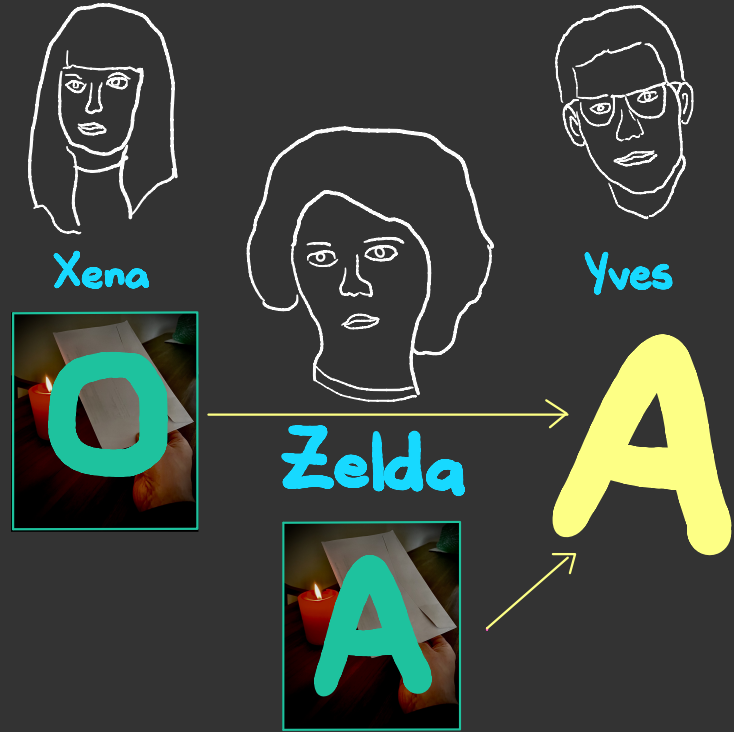
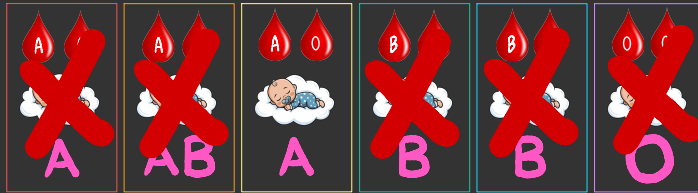
Zelda



Yves

e.g.

if Xena is type O,
then Yves must
be type A



the moral of the story

the value of information is conditional

~~don't trust hospitals~~



the value of information is **conditional**

without Zelda's bloodtype,
Xena's bloodtype provides **no**
information about Yves'



given Zelda's bloodtype,
Xena's bloodtype **could** provide
information about Yves'

